

Simplifying complex expressions

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Factoring polynomials

Factor the quadratic:

$$x^2 - 5x + 6$$

Solution:

Notice: $-5 = (-2) + (-3)$ and $+6 = (-2) \cdot (-3)$.

Recall **FOIL** rule:

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

Therefore set $a = -2$ and $b = -3$ and we have:

$$x^2 - 5x + 6 = (x - 2)(x - 3)$$

Expanding polynomials

Expand the quadratic:

$$(x + 2)(x - 3)$$

Solution:

Recall FOIL rule: $(x + a)(x + b) = x^2 + (a + b)x + ab$.

Use $a = +2$ and $b = -3$:

$$(x + 2)(x - 3) = x^2 - x - 6$$

Recognizing expansions

IMPORTANT. When you see these patterns, think of their partners!

Patterns:

$$A^2 - B^2, \quad x^2 - 9, \quad \frac{x^3 - y^3}{x - y}, \quad \begin{array}{c} a^2 + 2ab + b^2 \\ \text{-or-} \\ a^2 - 2ab + b^2 \end{array}$$

Partners:

$$A^2 - B^2 = (A - B)(A + B), \quad x^2 - 9 = (x - 3)(x + 3)$$

$$x^3 - y^3 = (x - y)(x^2 + xy + y^2) \quad \text{therefore} \quad \frac{x^3 - y^3}{x - y} = x^2 + xy + y^2$$

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

Factoring polynomials

Factor the cubic:

$$f(x) = x^3 + 3x^2 + 2x + 6$$

Solution:

First, *hunt for one root*. (Equivalently: Find one linear factor.)

Apply Rational Roots Theorem: consider all $\pm$ fractions using factors of the constant, 6. Start with integers.

E.g. -2 fails: $(-2)^3 + 3(-2)^2 + 2(-2) + 6 = +6 \neq 0$.

But -3 works: $(-3)^3 + 3(-3)^2 + 2(-3) + 6 = 0$. Therefore $(x - -3)$ is a factor, -3 is a root.

Using long division for $\frac{f(x)}{(x+3)}$, obtain $f(x) = (x + 3)(x^2 + 2)$.

It turns out $(x^2 + 2)$ is fully factored, so we are done.

Rational functions

Simplify the rational function:

$$f(x) = \frac{x^2 - 4}{x^2 - 2x}$$

Solution:

$$\frac{x^2 - 4}{x^2 - 2x} \xrightarrow{\text{factor}} \frac{(x - 2)(x + 2)}{x(x - 2)}$$

$\xrightarrow{\text{cancel}}$

$$f(x) = \frac{x + 2}{x}$$

(valid when $x \neq 2$)

Warning! Actually $f(2) = \text{undefined}$, so $f(2) \neq 2$.

Rational functions

Express this sum as a simple fraction:

$$\frac{2}{x} + \frac{3}{x+1}$$

Solution:

$$\begin{aligned}\frac{2}{x} + \frac{3}{x+1} &\ggg \frac{2}{x} \cdot \frac{(x+1)}{(x+1)} + \frac{3}{x+1} \cdot \frac{x}{x} \\ &\ggg \frac{2(x+1) + 3x}{x(x+1)} \\ &\ggg \frac{2x + 2 + 3x}{x(x+1)} \\ &\ggg \frac{5x + 2}{x(x+1)}\end{aligned}$$

Rational functions

Simplify this expression:

$$\frac{\frac{1}{x} + \frac{1}{y}}{\frac{1}{x} - \frac{1}{y}}$$

Solution:

$$\frac{\frac{1}{x} + \frac{1}{y}}{\frac{1}{x} - \frac{1}{y}} \gg \gg \frac{\frac{y+x}{xy}}{\frac{y-x}{xy}} \gg \gg \frac{y+x}{y-x}$$

Rational functions

Simplify the fraction:

$$\frac{x^2 - 9}{x^2 - 6x + 9}$$

Solution:

$$\frac{x^2 - 9}{x^2 - 6x + 9} \gg \gg \frac{(x - 3)(x + 3)}{(x - 3)^2}$$

$$\gg \gg \frac{x + 3}{x - 3} \quad (\text{valid when } x \neq 3)$$

Notice: in this case $x = 3$ is not in the original domain, so the simplified fraction is equally valid.

Log rules

Expand and simplify the expression using log rules:

$$\log(8x^3)$$

Solution: Recall the log rule:

“log converts *input products* to *output sums*”:

$$\log(AB) = \log(A) + \log(B)$$

And in consequence, the *power rule*:

$$\log(x^a) = a \log(x)$$

Therefore:

$$\log(8x^3) \gg \gg \log(8) + \log(x^3) \gg \gg \log(8) + 3 \log(x)$$

Log rules

Simplify the expression:

$$\log_2(32)$$

Solution: Recall the meaning of log with base b :

$$\log_b(x) = y \iff b^y = x$$

Now *notice*: $32 = 2^5$.

Therefore:

$$\log_2(32) = 5 \iff 2^5 = 32 \quad \checkmark$$

So $\log_2(32) = 5$.

Log rules

Expand the expression as a sum of logs of single factors:

$$\log\left(\frac{10x}{5y}\right)$$

Solution: Recall the log rule:

“log converts *input quotients* to *output differences*”:

$$\log(A/B) = \log(A) - \log(B)$$

Therefore:

$$\log\left(\frac{10x}{5y}\right) \gg \gg \log(10x) - \log(5y)$$

$$\gg \gg \log(10) + \log(x) - \log(5) - \log(y)$$

Log rules

Simplify:

$$\frac{\ln 125}{\ln 25}$$

Solution: Recall the log rule:

“log converts *input powers* to *output factors*”:

$$\log(A^b) = b \cdot \log(A)$$

Remember!: $\ln x = \log_e x$

Therefore:

$$\begin{aligned} \frac{\ln 125}{\ln 25} &\ggg \frac{\ln 5^3}{\ln 5^2} \\ &\ggg \frac{3 \ln 5}{2 \ln 5} \quad \text{cancel} \quad \frac{3}{2} \end{aligned}$$

Exponent rules

Simplify:

$$(3x)^2, \quad 2^x \cdot 4^x, \quad \frac{5^{x+2}}{5^{x-1}}$$

Solution: Recall the power rule:

$$A^b \cdot A^c = A^{b+c}$$

Therefore:

$$(3x)^2 \gggg 9x^2$$

$$2^x \cdot 4^x \gggg 2^x \cdot 2^{2x} \gggg 2^{3x} \gggg 8^x$$

$$\frac{5^{x+2}}{5^{x-1}} \gggg 5^{x+2} \cdot 5^{-(x-1)} \gggg 5^3 = 125$$

Log / Exp simplification

Simplify:

$$e^{-4 \ln(10)}, \quad \log_2(36) - \log_2(6), \quad \log(3) - \log(10^z)$$

Solution:

$$e^{-4 \ln(10)} \ggg e^{\ln(10^{-4})} \ggg 10^{-4}$$

$$\log_2(36) - \log_2(6) \ggg \log_2(36/6) \ggg \log_2(6)$$

$$\log(3) - \log(10^z) \ggg \log(3 \cdot 10^{-z}) \ggg \log(10^{\log 3} \cdot 10^{-z})$$

$$\ggg \log(10^{(\log 3) - z}) \ggg (\log 3) - z$$

Log / Exp inversion

For what x values are these equalities valid?

$$e^{\ln x} \underset{(1)}{=} x, \quad \ln(e^x) \underset{(2)}{=} x, \quad 10^{\log_{10} x} \underset{(3)}{=} x, \quad \log_{10}(10^x) \underset{(4)}{=} x$$

Solution:

Exponentials have *any input BUT positive output*

Logs reverse this:

only positive input allowed BUT any output is possible

So, equations (1) and (3) hold for $x > 0$ only.

Whereas equations (2) and (4) hold for *all* x .

Trig identities

Simplify:

$$\sqrt{1 - \sin^2 \theta}$$

Solution: Recall the fundamental trig identity:

$$\cos^2 \theta + \sin^2 \theta = 1$$

Therefore:

$$1 - \sin^2 \theta = \cos^2 \theta$$

and thus:

$$\sqrt{1 - \sin^2 \theta} \gggg \sqrt{\cos^2 \theta} \gggg |\cos \theta|$$

Trig identities

Simplify:

$$\frac{\sin^2 x}{\sec^2 x - 1}$$

Solution: Recall the second fundamental trig identity:

$$1 + \tan^2 \theta = \sec^2 \theta$$

Therefore:

$$\sec^2 x - 1 = \tan^2 x$$

and thus:

$$\frac{\sin^2 x}{\sec^2 x - 1} \ggg \frac{\sin^2 x}{\tan^2 x} \ggg \sin^2 x \cdot \frac{\cos^2 x}{\sin^2 x} \ggg \cos^2 x$$

Trig inversion

Which of these is true for all positive x ?

$$(1) \sin(\sin^{-1}(x)) = x, \quad (2) \sin^{-1}(\sin(x)) = x$$

Solution: Neither!

Recall the precise *domain* and *range* of $\sin^{-1}(x)$:

$$\sin^{-1} : [-1, 1] \longrightarrow [-\pi/2, \pi/2]$$

When $x = 10$, equation (1) doesn't make *sense* on the left side.

Equation (2) makes sense at $x = 10$, and $\sin(10)$ can be computed with result in $[-1, 1]$. However, the output of $\sin^{-1}$ will (by definition) be placed in $[-\pi/2, \pi/2]$, so it could never equal 10.

Addendum:

(1) holds for $x \in [-1, 1]$. (2) holds for $x \in [-\pi/2, \pi/2]$.

Q: For what x is $\tan(\tan^{-1}(x)) = x$ and $\cos(\cos^{-1}(x)) = x$?

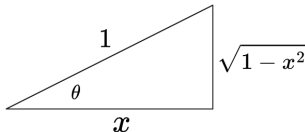
Trig simplification

Simplify the trig expression using a triangle:

$$\tan(\cos^{-1}(x))$$

Solution:

Set $\theta = \cos^{-1}(x)$, and therefore $\cos(\theta) = \frac{x}{1}$. Draw the triangle:



Using this triangle, we see that $\tan(\theta) = \frac{\sqrt{1-x^2}}{x}$